Private Reserve · DIN controller

Wiring Guide

Boards A, B, and C as built: earth-side control, floating gate driver, and HV switch stage.

Screw-terminal IDs on these pages are the names used on the operator screens.

1. Board A — Earth-side control
2. Board B — Floating gate driver
3. Board C — HV switch stage

Board A — Earth-side control

Dual Adafruit Perma-Proto 1/2 carrier for the LilyGo T-Display-S3 and the EC11 menu encoder. Field I/O and power leave through edge screw terminals. This board is earth-referenced only.

Draft. Header pin order along each edge is taken from the carrier with the display up and USB-C on the left. The LilyGo pinmap photo is the module bottom, so its LEFT/RIGHT labels are mirrored here. Screw terminals are numbered **T1–T24** left to right on each edge (T1 nearest USB). Outer rails are as-built. Inner rails are proposed. Column span 7–18 is assumed. On-board jumper paths are documented elsewhere.

No floating copper. Do not land `+5V_ISO`, `+12V_GD`, `−70V`, `+70V`, or `SW` on Board A. `MCU_GND` is not `−70V`.

1. Placement

View with the display facing you. USB-C is on the left. The EC11 is on the right. Pin 12 of each header is nearest USB. Screw terminals are numbered **T1–T12** on the top edge and **T13–T24** on the bottom edge, left to right. The **top** edge starts at T1 (`3V`, `GND`, one NC, then PWM, DIR, limits, I^{2}C). The **bottom** edge starts at T13 (`5V`, `GND`, five NC including T17–T19, then OPEN, LIM-L, ISENSE, HALL, 3V). The EC11 uses GPIO11–13 on the T-Display header directly; see EC11 on-board wiring.

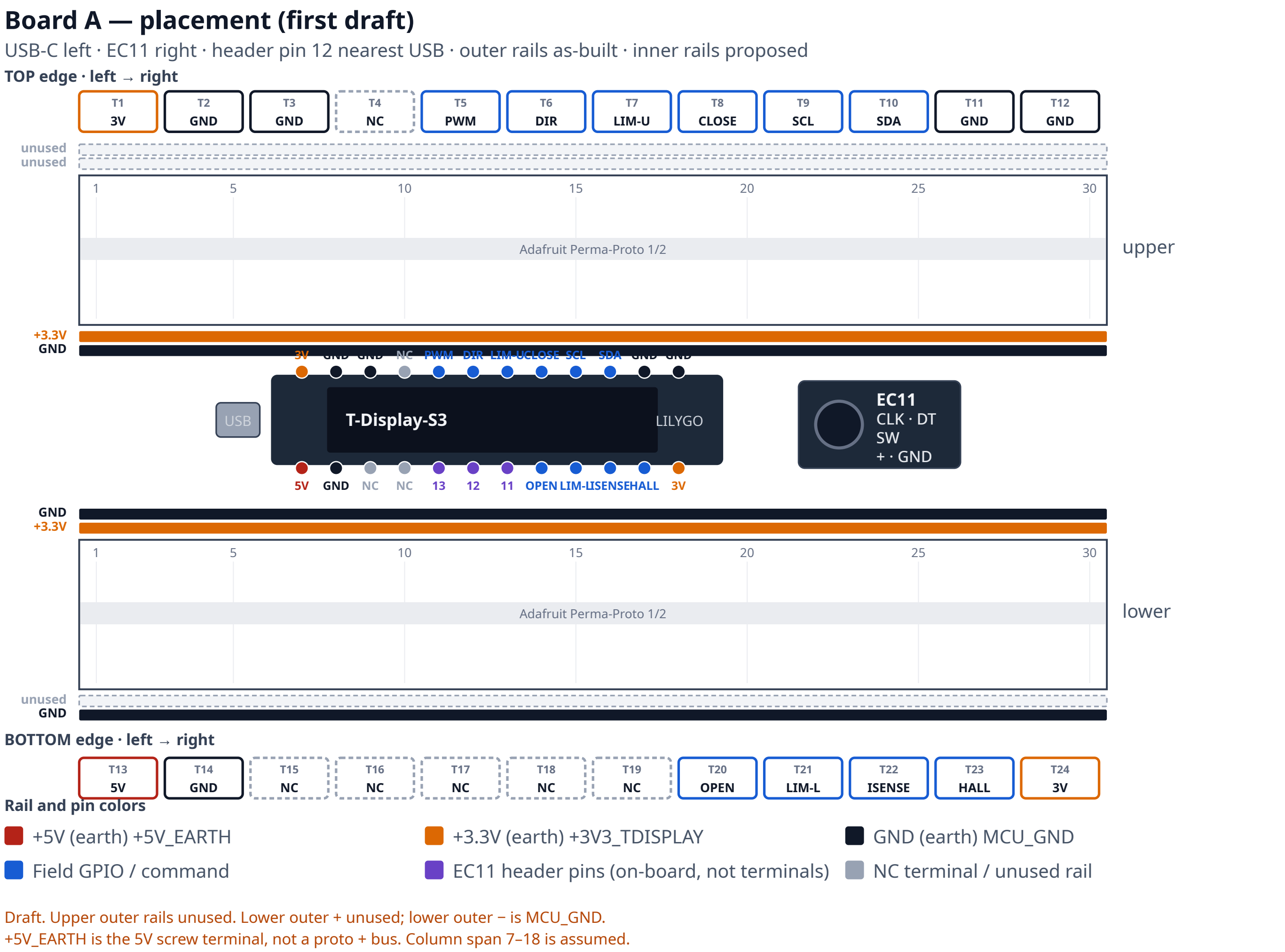


Figure 1. Grid-true placement. Terminal IDs T1–T24 label each screw block left to right. Dashed gray bars are unused rails (as-built: both upper outer buses, and the lower outer +). Purple dots are T-Display header pins wired to the on-board EC11 (GPIO11–13). T17–T19 are NC; the encoder does not use those terminals.

2. Rails

Each half has four full-width buses. On the as-built unit both **upper outer** rails are unused, and the **lower outer +** rail is unused. The **lower outer −** rail is `MCU_GND`. Inner-rail nets below are still proposed. `+5V_EARTH` lands on the **5V** screw terminal and the T-Display **5V** pin; it is not on a proto + bus. Sensors / EC11 + should see `+3V3_TDISPLAY` from the module **3V** pin. Do not back-feed 3.3 V into the module.

Rail	Net	Diagram color
Upper outer −	unused (as-built; no jumper)	■ unused
Upper outer +	unused (as-built; no jumper)	■ unused
Upper inner +	`+3V3_TDISPLAY` (proposed)	■ +3.3V (earth)
Upper inner −	`MCU_GND` (proposed)	■ GND (earth)
Lower inner −	`MCU_GND` (proposed)	■ GND (earth)
Lower inner +	`+3V3_TDISPLAY` (proposed)	■ +3.3V (earth)
Lower outer +	unused (as-built; no jumper)	■ unused
Lower outer −	`MCU_GND` (as-built)	■ GND (earth)

3. What attaches

Power

Board A runs from the earth-side Mean Well **HDR-15-5 #1** supply (`+5V_EARTH` / `MCU_GND`). Land the PSU positive on **T13** (`5V`) and the PSU negative on **T14** (`GND`). Those terminals feed the T-Display **5V** pin and the board `MCU_GND` reference. USB-C on the module remains for bench programming. Do not back-feed 3.3 V into the module **3V** pin.

Screw terminals

Numbering follows Figure 1: left to right with the display facing you and USB-C on the left. T1 and T13 are nearest USB.

Top edge — T1–T12

Terminal	Label	GPIO / net	Attaches
T1	**3V**	`+3V3_TDISPLAY`	SS49E VCC, AS5600 VCC (from module **3V**)
T2	**GND**	`MCU_GND`	Sensor, limit, and rocker returns
T3	**GND**	`MCU_GND`	Sensor, limit, and rocker returns. Lands **R6** (PWM) and **R7** (DIR) 10 kΩ pull-downs
T4	**NC**	—	Module NC. Unassigned. May be jumpered to `MCU_GND`.
T5	**PWM**	GPIO16	Board B 6N137 LED. On-board **R6** 10 kΩ to T3 `MCU_GND`
T6	**DIR**	GPIO21	Board B U4 LED. On-board **R7** 10 kΩ to T3 `MCU_GND`
T7	**LIM-U**	GPIO17	Upper NC limit to GND
T8	**CLOSE**	GPIO18	Close rocker (active-low, paralleled pair)
T9	**SCL**	GPIO44	AS5600 SCL
T10	**SDA**	GPIO43	AS5600 SDA
T11	**GND**	`MCU_GND`	Sensor, limit, and rocker returns; Board B LED return
T12	**GND**	`MCU_GND`	Sensor, limit, and rocker returns

Bottom edge — T13–T24

Terminal	Label	GPIO / net	Attaches
T13	**5V**	`+5V_EARTH`	**Power in.** Mean Well HDR-15-5 #1 (+) → T-Display **5V**
T14	**GND**	`MCU_GND`	**Power in.** Mean Well HDR-15-5 #1 (−) / earth-side return
T15	**NC**	—	Module NC. Unassigned.
T16	**NC**	—	Module NC. Unassigned.
T17	**NC**	—	Unassigned. Same index as GPIO13; EC11 SW is on-board from the header (EC11 wiring).
T18	**NC**	—	Unassigned. Same index as GPIO12; EC11 DT is on-board from the header.
T19	**NC**	—	Unassigned. Same index as GPIO11; EC11 CLK is on-board from the header.
T20	**OPEN**	GPIO10	Open rocker (active-low, paralleled pair)
T21	**LIM-L**	GPIO3	Lower NC limit to GND
T22	**ISENSE**	GPIO2	ACS712-05B OUT after 10 kΩ/20 kΩ (≤ 3.3 V)
T23	**HALL**	GPIO1	SS49E OUT
T24	**3V**	`+3V3_TDISPLAY`	SS49E VCC, AS5600 VCC (from module **3V**)

EC11 (on-board)

The menu encoder wires directly to T-Display-S3 header pins. It does not use screw terminals. Terminals **T17–T19** are NC on the as-built unit.

EC11 breakout	T-Display pin	GPIO / net
CLK	LEFT GPIO11	GPIO11
DT	LEFT GPIO12	GPIO12
SW	LEFT GPIO13	GPIO13
+	**3V**	`+3V3_TDISPLAY`
GND	**GND**	`MCU_GND`

Full notes, terminal clarification, and links: EC11 on-board wiring.

On-board pull-downs

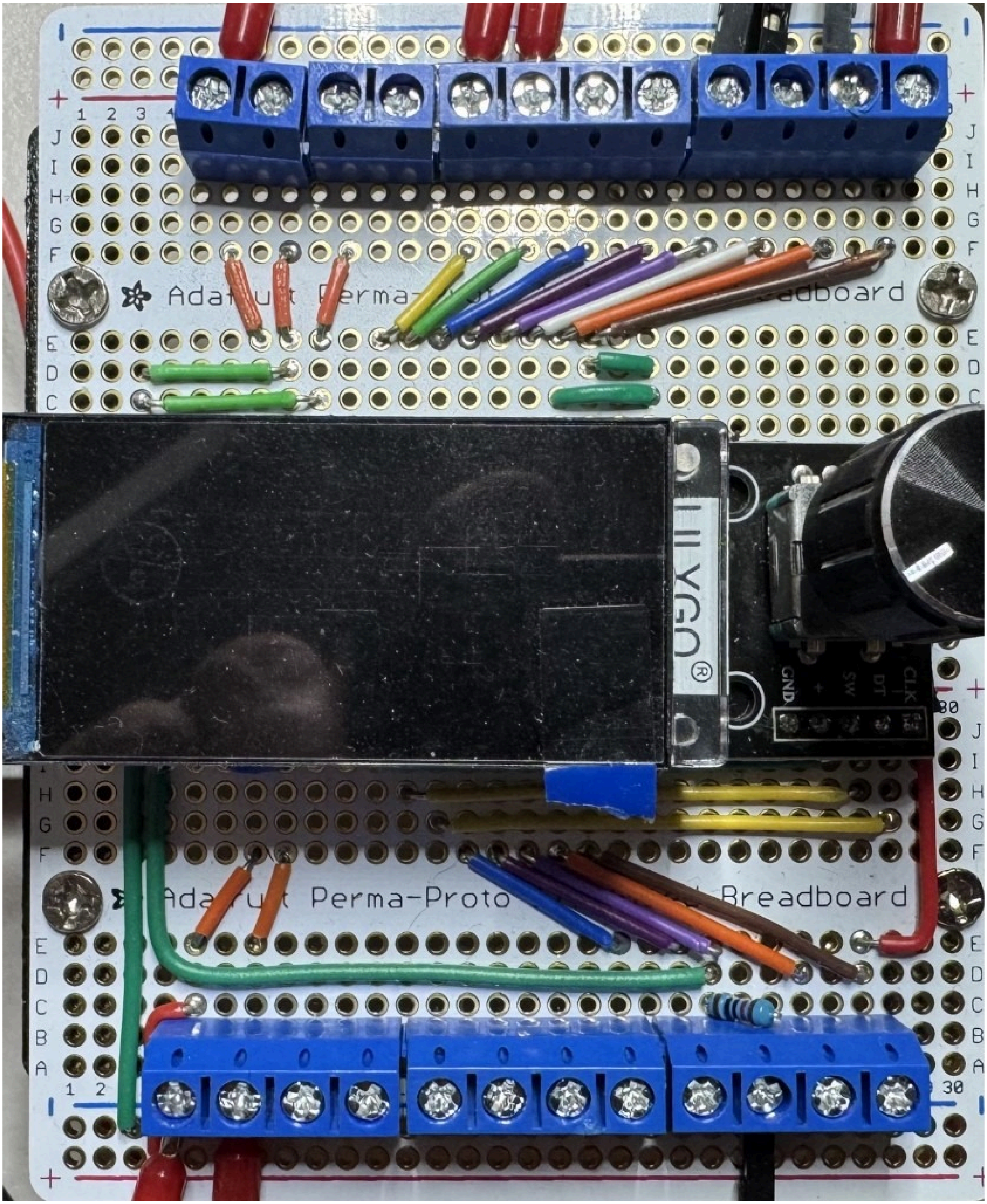
R6 (10 kΩ) sits between **T5** (`PWM`, GPIO16) and **T3** (`MCU_GND`). **R7** (10 kΩ) sits between **T6** (`DIR`, GPIO21) and the same T3 return. They hold the Board B optocoupler LEDs off if the T-Display is unpowered or resetting. Series LED resistors stay on Board B.

Board A → Board B

Keyed 3-pin earth-side cable from **T5** (`PWM`), **T6** (`DIR`), and a `MCU_GND` terminal (for example **T11** or **T14**). LED resistors stay on Board B.

4. As-built photo

Line art is the labeled figure. The photo shows the unit that was built.



Wired carrier. USB-C on the left, EC11 on the right.

Board B — Floating gate driver

Single Adafruit Perma-Proto 1/2 carrier for the 6N137 / 4N35 optocouplers, Q2 fail-off, TC4426 gate driver, and Q3 K1 coil switch. Earth-side LED inputs occupy columns 1–5; the floating gate-driver region occupies columns 6–30. Floating power and inter-board links leave through screw terminals on the board edges.

Draft. View with U1 (6N137) on the left and U4 (4N35) on the right, matching the as-built photo. Vertical stacks (**T1–T3** upper left, **T6–T7** lower left) are numbered top → bottom. Horizontal screws (**T4–T5** top edge) are numbered left → right. **T1–T11** are as-built on the populated unit. Float rails: top **+** is `+12V_GD`, bottom **+** is `+5V_ISO`, both **−** rails share `−70V`. On-board jumper paths and pin-by-pin IC wiring stay out of scope here; see Motor_Driver_Wiring.md.

Two galvanic domains. Columns 1–5 are the earth LED island: `PWM`, `DIR`, and `MCU_GND` from Board A (T-Display-S3 GPIO16 / GPIO21 and earth return). Those nets are earth-referenced only. Columns 6–30 are floating, referenced to `−70V`. U1 and U4 translate the command signals optically; no copper joins the domains. Never tie `MCU_GND` to `−70V`. After HDR-15-5 #2 and HDR-15-12 negatives bond to `−70V`, treat both floating supplies as hazardous relative to earth. Do not clip an earth-referenced scope ground to `−70V`, the gate node, or `SW`.

1. Placement

View the board from above with U1 on the left and U4 on the right, as in the as-built photo. U3 (TC4426) sits in the center gutter; Q2 is beside U3; Q3 is beside U4. The isolation barrier falls at approximately column 5: a physical break in the proto buses separates the earth LED island from the floating region. The upper-left corner holds a vertical 3-position block (**T1–T3**). The earth-side Board A cable lands on **T4** (`PWM`) and **T5** (`MCU_GND`). The lower-left stack (**T6–T7**) carries HDR-15-5 #2. The upper-right block (**T8–T9**) takes `DIR` from Board A and drives PWM out to Board C. The lower-right block (**T10–T11**) wires the Phoenix K1 coil.

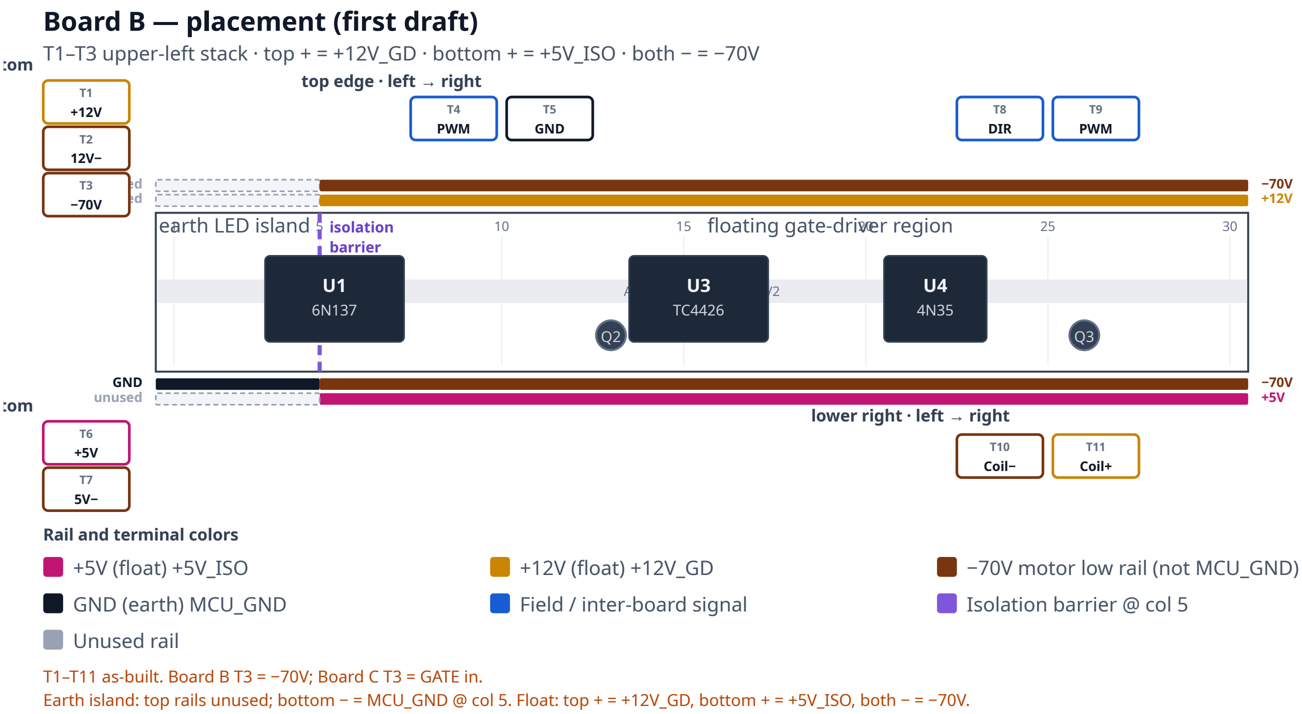


Figure 1. Grid-true placement draft. Left-edge stacks (**T1–T3**, **T6–T7**) sit beside the Perma-Proto outline, matching the physical board. Purple dashed line at column 5 is the isolation barrier. LED resistors, Q2 bias, TC4426 decoupling, and the 100 Ω gate resistor are on-board only.

2. Rails

Each Perma-Proto 1/2 has four full-width buses (top −, top +, bottom −, bottom +). A single bus cannot carry two domains; the isolation barrier at column 5 keeps earth-side copper out of the floating region. On the as-built unit, the bottom **−** rail on the earth island (column 5) lands `MCU_GND`. In the floating region, both **−** rails are `-70V`, the top **+** rail is `+12V_GD`, and the bottom **+** rail is `+5V_ISO`.

Rail (U1 left, U4 right)	Earth island (cols 1–5)	Floating region (cols 6–30)	Diagram color
Top −	unused	`-70V` (as-built)	■ −70V
Top +	unused	`+12V_GD` (as-built)	■ +12V_GD
Bottom −	`MCU_GND` (as-built @ col 5)	`-70V` (as-built)	■ −70V / ■ GND
Bottom +	unused	`+5V_ISO` (as-built)	■ +5V_ISO

3. What attaches

Power

Board B draws from two floating DIN supplies whose negatives are bonded only to `-70V`. The command cable from Board A carries no floating power — only earth-referenced GPIO and `MCU_GND`.

Land Mean Well **HDR-15-12** on the upper-left stack: positive on **T1** (`+12V_GD`) and negative on **T2** (12 V supply return — bonded to `−70V`, not `MCU_GND`). That feed drives the top **+** rail (U3 VDD, Q2 bias, K1 coil branch).

Land Mean Well **HDR-15-5 #2** on the lower-left stack: positive on **T6** (`+5V_ISO`, top screw) and negative on **T7** (bottom screw — floating return bonded to `−70V`, not `MCU_GND`) → bottom **+** rail → U1 VCC and VE. Do not connect either floating supply negative to `MCU_GND` or protective earth.

T3 (Board B) is the `−70V` landing for the short harness to Board C (Q1 source / U3 GND return). Do not confuse it with Board C **T3**, which receives PWM from Board B **T9**.

Screw terminals

Numbering follows Figure 1. Vertical stacks are top → bottom; horizontal screws are left → right on that edge.

Upper left — T1–T3 (top → bottom, as-built)

Terminal	Label	Net	Attaches
T1	**+12V**	`+12V_GD`	**Power in.** Mean Well HDR-15-12 (+) → top **+** rail
T2	**12V−**	`−70V`	**Power in.** Mean Well HDR-15-12 (−). Floating supply return bonded to `−70V`. This is not `MCU_GND`.
T3	**−70V**	`−70V`	Board B → Board C shared motor low rail (Q1 source / U3 pin 3). Keep this link short.

Top edge — T4–T5 (left → right, as-built)

Terminal	Label	Net	Attaches
T4	**PWM**	`PWM` (GPIO16)	Board A **T5** ← T-Display-S3 GPIO16. Earth island only → U1 LED through 220 Ω on-board. Optically isolated to the floating domain.
T5	**GND**	`MCU_GND`	Board A **T12** ← T-Display earth return. U1 / U4 LED cathodes. Never tie to `−70V` or floating supplies.

Lower left — T6–T7 (top → bottom, as-built)

Terminal	Label	Net	Attaches
T6	**+5V**	`+5V_ISO`	**Power in.** Mean Well HDR-15-5 #2 (+) → bottom **+** rail → U1 VCC / VE
T7	**5V−**	`−70V`	**Power in.** Mean Well HDR-15-5 #2 (−). Floating supply return bonded to `−70V`. Not `MCU_GND`.

Upper right — T8–T9 (left → right, as-built)

Terminal	Label	Net	Attaches
T8	**DIR**	`DIR` (GPIO21)	Board A **T6** ← T-Display-S3 GPIO21. Earth island → U4 LED (270 Ω on-board). LED return shares **T5** `MCU_GND` — no separate earth screw for DIR. Optically isolated to Q3 / K1 coil.
T9	**PWM out**	`PWM` / gate	TC4426 OUT B through 100 Ω (on-board) → Board C **T3**. Floating domain. Return via Board B **T3** `−70V`. Keep harness short.

Lower right — T10–T11 (left → right, as-built)

Terminal	Label	Net	Attaches
T10	**Coil−**	K1 coil (−) / Q3 drain	Phoenix **PLC-RSC-12DC/21-21** coil (−) via field wire. Low side switched by Q3 to `−70V` on-board.
T11	**Coil+**	`+12V_GD`	Phoenix K1 coil (+) from `+12V_GD` rail (HDR-15-12 / top **+** bus). D5 flyback (1N4007) is installed across these terminals on the as-built unit; cathode/band toward **T11**.

K1 coil (on-board driver, DIN relay)

Q3 switches the Phoenix K1 coil low-side on the floating domain. Field wires to the DIN relay coil land on **T10** (Coil−) and **T11** (Coil+). Do not route coil current through the Board B → Board C PWM link (**T9**).

Signal	Net	Attaches
K1 coil +	`+12V_GD`	**T11** → Phoenix PLC-RSC-12DC/21-21 coil (+)
K1 coil −	Q3 drain	**T10** ← relay coil (−); Q3 source → `−70V`
PWM command	`PWM`	T-Display-S3 GPIO16 → Board A **T5** → Board B **T4** → U1 LED (220 Ω on-board); optically isolated to Q2 / U3 / gate
Direction command	`DIR`	T-Display-S3 GPIO21 → Board A **T6** → Board B **T8** → U4 LED; return via **T5** `MCU_GND`

Coil and contact detail: K1 floating coil option.

On-board only (not on screw terminals)

These subsystems stay on the Perma-Proto and are wired per Motor_Driver_Wiring.md:

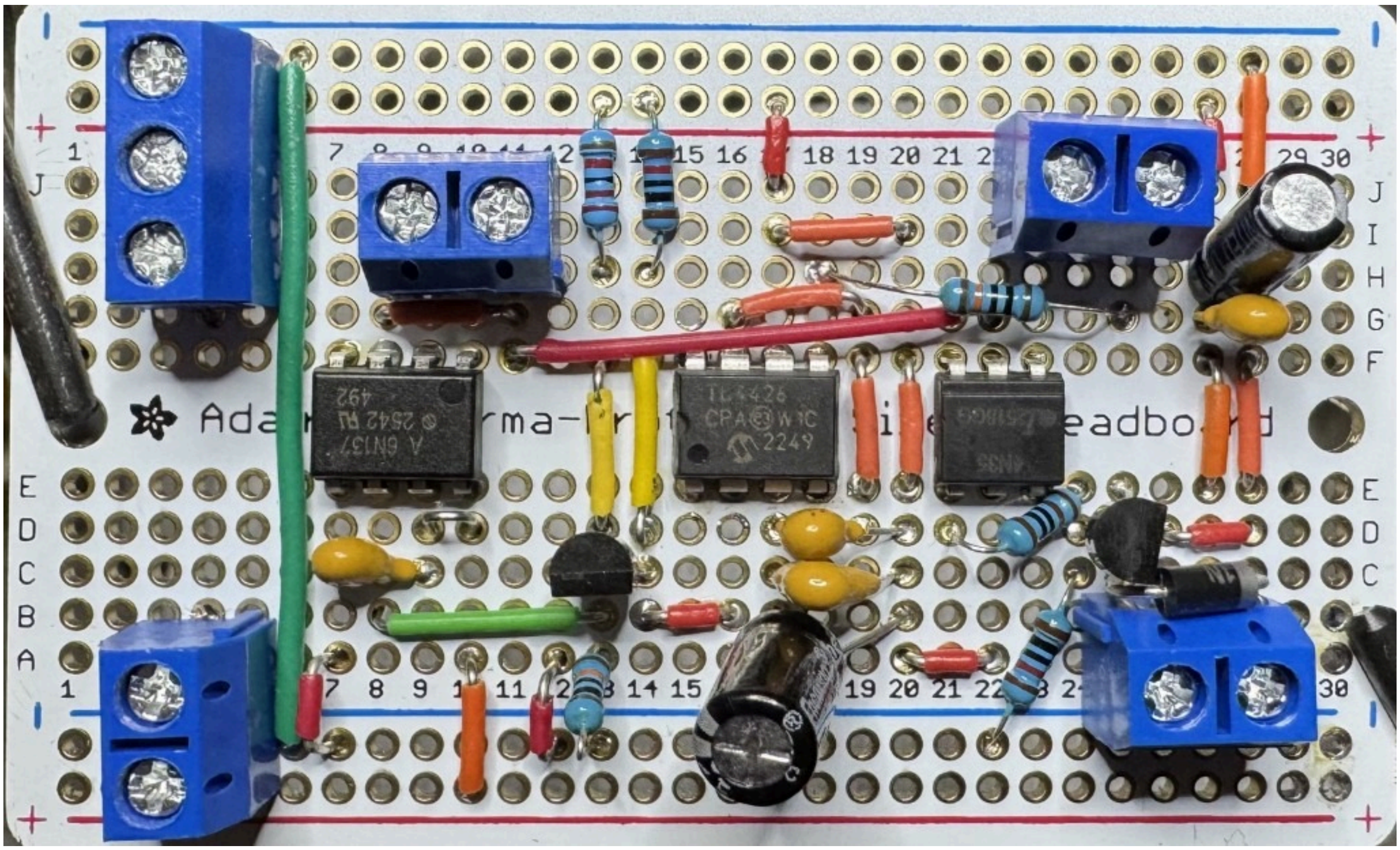
- U1 6N137 — PWM isolation; pin 6 → Q2 base
- Q2 PN2222A — fail-off inverter (Rb2, Rbe2, Rc2 on-board)
- U3 TC4426 — dual inverting gate driver; OUT B → 100 Ω → Board C gate path
- U4 4N35 + Q3 2N7000TA — direction opto and coil low-side switch
- 220 Ω (U1 LED) and 270 Ω (U4 LED) beside the optocouplers
- U3 decoupling: 100 nF, 1 μF across pins 6–3; 47 μF bulk nearby

Inter-board cables

Cable	Board B landing	Domain
Board A → Board B	**T4** (`PWM` in), **T5** (`MCU_GND`), **T8** (`DIR`)	Earth only; translated optically by U1 / U4
Board B → Board C	**T3** (`−70V`) + **T9** (PWM out → Board C **T3**)	Floating; keep short
Board B → DIN K1 coil	**T10** (Coil−), **T11** (Coil+ / `+12V_GD`)	Floating

4. As-built photo

Line art is the labeled figure. The photo shows the unit that was built.



Populated carrier. U1 (6N137) on the left, U3 (TC4426) in the center, U4 (4N35) on the right. Isolation barrier at column 5.

Board C — HV switch stage

Adafruit Perma-Proto **1/4** (columns 1–15) carrying Q1 IRFP460, D9 STTH8R06 flyback, TVS1 P6KE200A, and the local `SW` / flyback loop. All copper is in the floating HV domain referenced to `−70V`. Screw terminals **T4** and **T5** wire to the DIN K1 return contacts; motor leads land on K1 COM, not on this board.

Draft. View with the screw terminals on the **bottom** edge and the IRFP460 heatsink at the **top**. Terminals are numbered **T1–T5** left to right (T1 at column 3). On the as-built unit all four Perma-Proto power rails are **unused** — HV nets use column jumpers, not the silkscreen ± buses. On-board gate clamp parts (10 kΩ, BZX85C15) and row-column routing are documented elsewhere. Pin-by-pin IC wiring is not duplicated here.

Floating HV — up to 140 VDC. Every net on Board C is hazardous relative to earth when the motor supply is energized. `−70V` is **not** `MCU_GND`. Do not bond this board to protective earth except through an analyzed heatsink isolator. Q1 tab and drain are live at `SW`. Use an isolated or differential probe referenced to `−70V`, not earth ground. Discharge the bus and verify safe voltage before service.

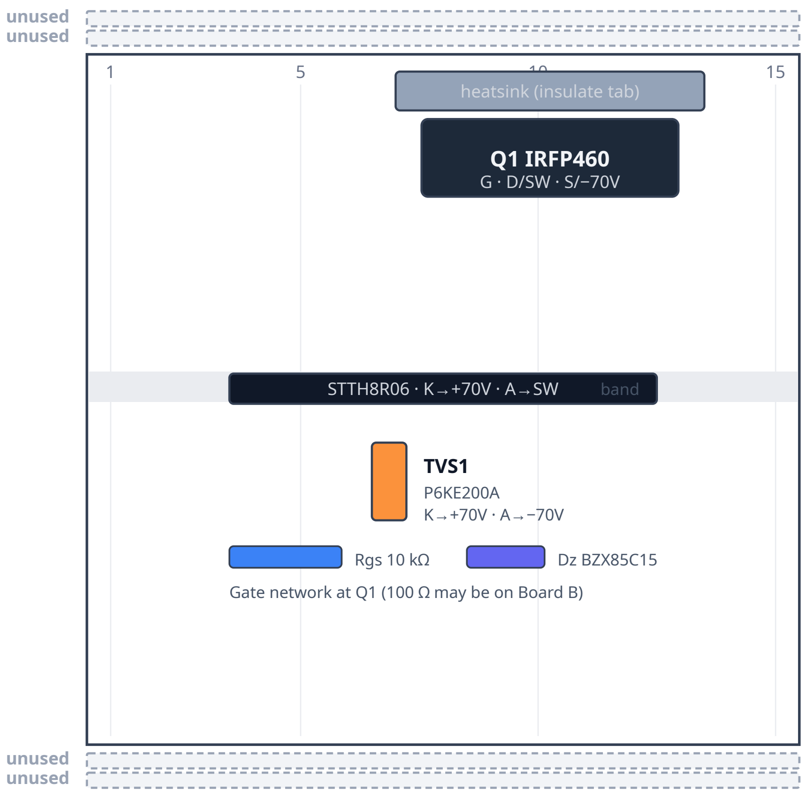
1. Placement

Mount on DIN rail with ≥20 mm clearance beside the heatsink for airflow (see DIN layout). Q1 sits under an insulated heatsink at the top of the proto. D9 spans the mid board with cathode toward the fused `+70V` side and anode to `SW`. TVS1 clamps between the on-board `+70V` and `−70V` nodes (via column wiring, not the Perma-Proto rails). Five screw terminals on the bottom edge accept field and inter-board cables; there are no terminals on the top edge.

Flyback loop constraint. When Q1 turns off, motor inductance pushes current through D9 from `SW` to `+70V`. That path — `SW` → D9 anode → D9 cathode → `+70V` — must stay physically short and entirely on Board C. Figure 1 shows this as a dashed callout box below the proto; it is a *placement rule*, not a specific jumper wire to install. Do not run the flyback loop through a long cable to Board B or the DIN rail.

Board C — placement (first draft)

Heatsink up · terminals down · floating HV only · cols 1–15



Flyback loop

Placement rule — not a wire path

SW → **D9** → **+70V**

When Q1 turns off, motor inductance pushes current through D9.

Path: SW → D9 anode → D9 cathode → +70V.

Keep D9 physically close to the SW node and +70V landing.

Do not route this flyback loop off Board C.

BOTTOM edge · left → right

T1	T2	T3	T4	T5
+70V	**−70V**	**GATE**	**NC1**	**NO2**

Net colors

- +70V fused high rail
- SW — K1 NC1 / NO2 (T4 / T5)
- Dashed box = placement rule (not a net)
- −70V motor low / Q1 source
- GATE from Board B TC4426
- Unused Perma-Proto rail

Draft. All four Perma-Proto rails are unused on the as-built unit. T4/T5 land K1 return contacts (NC1/NO2); motor leads are on K1 COM.
As-built: T1/T2/T4/T5 are reinforced for motor current; T3 is gate only.
Never join MCU_GND to −70V. Q1 tab is live at SW — insulate the heatsink.

Figure 1. Grid-true placement (columns 1–15). Terminal IDs T1–T5 label each screw block left to right. Dashed gray bars are unused Perma-Proto rails. T4 and T5 are separate K1 return-contact landings (NC1 / NO2), both on the `SW` net on-

board. The full-width dashed amber box below the proto is a placement constraint for the flyback path — not a wire diagram.

2. Rails

The 1/4 Perma-Proto has four full-width buses (15 columns). On the as-built unit **all four rails are unused** — no jumpers connect them to `+70V`, `−70V`, or any other net. HV distribution uses column strips and point-to-point jumpers instead. Do not land `MCU_GND`, `+5V_EARTH`, `+5V_ISO`, or `+12V_GD` on any Board C bus.

Rail	Net	Diagram color
Upper outer −	unused (as-built; no jumper)	■ unused
Upper inner +	unused (as-built; no jumper)	■ unused
Lower inner −	unused (as-built; no jumper)	■ unused
Lower outer +	unused (as-built; no jumper)	■ unused

3. What attaches

Power

Board C receives the motor outer rails from the inter-box 140 V feed and Wago distribution (see DIN layout, row B). AnTek `+70V` is interlocked in the **PSU box** (AC-in and DC-out relays, coils on `+12V_GD`) before that feed reaches F1, the Wago star, K1, and **T1**. Run those coil leads in the **AC** inter-box passage, not beside the ±70 V PWM pair. Details: DIN enclosure — 140 V interlock relays. The fused `+70V` high rail enters at **T1** after that interlock and F1. The low outer rail `−70V` enters at **T2**. Those terminals feed D9 cathode, TVS1, and Q1 source on-board. K1 high-side contacts (NO1/NC2) take fused `+70V` from the Wago star, not through a second Board C terminal, unless your harness branches from T1.

Screw terminals

Numbering follows Figure 1: left to right with the heatsink up and terminals down. T1 is leftmost (column 3 on the as-built unit). On the as-built board, leads at the motor-current terminals (**T1**, **T2**, **T4**, **T5**) are **reinforced** for the higher continuous and PWM motor current. Use heavy-gauge, HV-rated wire on those paths; keep **T3** (gate) as a low-current signal lead. See DIN layout for 140 V path gauge guidance (18 AWG minimum; 16 AWG preferred for runs >1 m).

Bottom edge — T1–T5

Terminal	Label	Net	Attaches
T1	**+70V**	`+70V`	**Power in.** After PSU-box DC interlock, then F1 / Wago → D9 K, TVS1 K (board-internal)
T2	**−70V**	`−70V`	**Power in.** Motor low outer rail → Q1 source, TVS1 A, Rgs/Dz return (board-internal)
T3	**GATE**	gate	Board B TC4426 OUT B (100 Ω may be on B or at Q1; see Motor_Driver_Wiring.md)
T4	**NC1**	`SW`	K1 pin 12 (NC1) — return contact → Q1 drain / D9 anode on-board
T5	**NO2**	`SW`	K1 pin 24 (NO2) — return contact → same `SW` node as T4 on-board

On-board only (not on terminals)

Ref	Part	Notes
Q1	IRFP460NPBF	Low-side switch; tab = drain = `SW`
D9	STTH8R06D	Motor PWM flyback; cathode → `+70V`, anode → `SW`
TVS1	P6KE200A	Across `+70V` and `−70V` on-board (column wiring; rails unused)
Rgs	10 kΩ	Gate-to-source at Q1
Dz	BZX85C15	15 V gate clamp at Q1

Full pin tables: Motor_Driver_Wiring.md.

Board B → Board C

Keyed two-conductor floating cable from Board B **T9** (PWM out / gate) and **T3** (`−70V` gate-driver return / Q1 source reference) to Board C **T3** (GATE) and **T2** (`−70V`). Keep this harness short; do not route the D9 flyback loop off Board C.

Board C → DIN K1 (contacts)

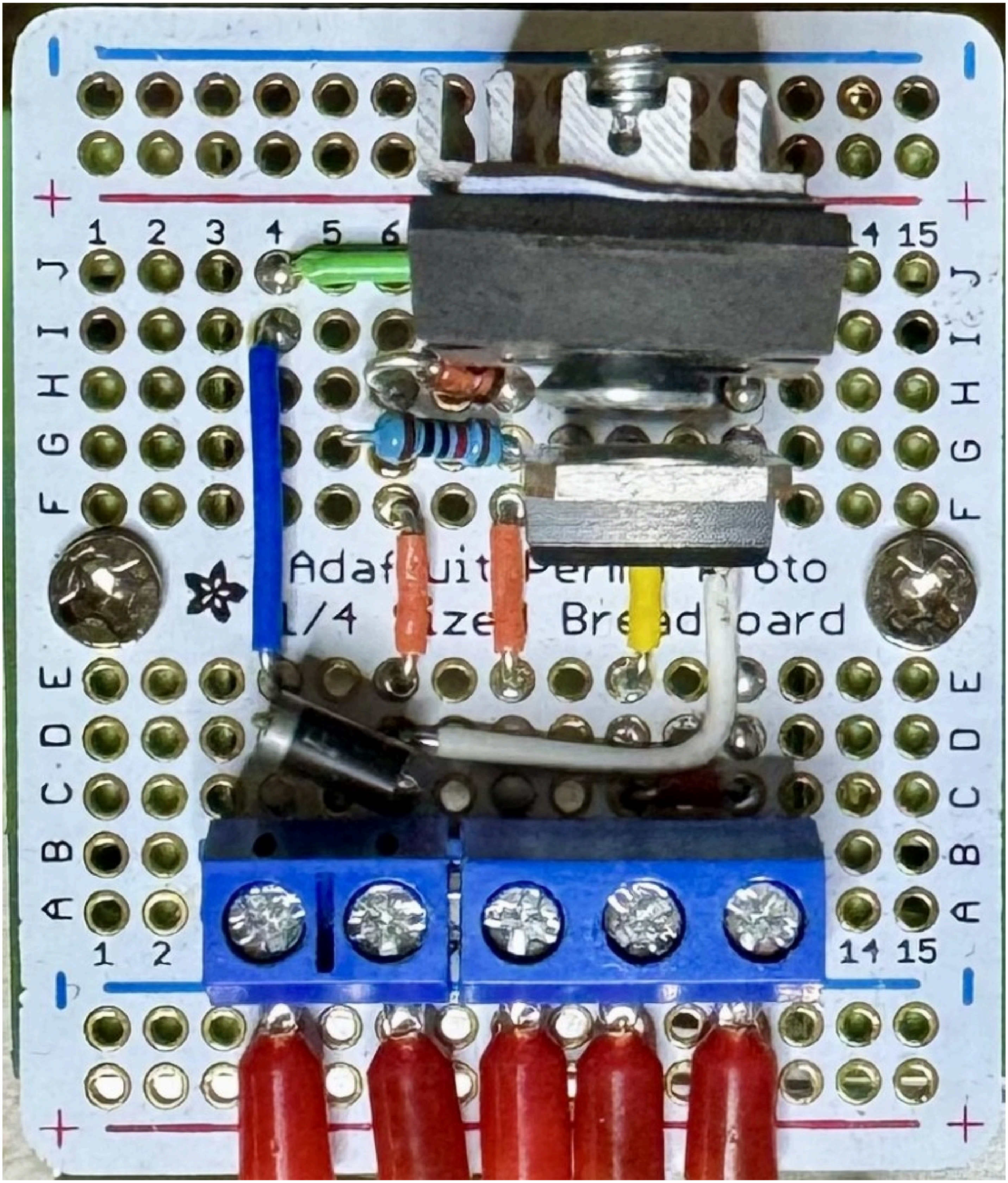
Direction switching uses the Phoenix K1 DPDT on the DIN rail. Board C lands the return-contact harness; high-side contacts and motor commons wire at the relay.

K1 pin	Net	From
NO1, NC2	`+70V` (fused)	Wago / F1 star (not a Board C terminal unless branched from T1)
NC1 (12)	`SW`	**T4**
NO2 (24)	`SW`	**T5**
COM1, COM2	`MOTOR_A`, `MOTOR_B`	Field motor harness (HV gland)

K1 coil (`+12V_GD` / Q3) is driven from Board B; see K1 floating coil option.

4. As-built photo

Line art is the labeled figure. The photo shows the unit that was built.



Wired carrier. Heatsink up, five bottom-edge terminals, columns 1–15. Red leads at T1–T2 and T4–T5 are reinforced for motor current; T3 (gate) is a low-current signal.